	Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current	762 - flash_fwd_kernel	(128, 1, 1)x(128, 1, 1)	3.36 ms	4,685,154	0 - NVIDIA GeForce RTX 3090	1.39 Ghz	[123817] python3.12	

GPU Speed Of Light Throughput

All

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	34.95	Duration [ms]	3.36
Memory Throughput [%]	15.20	Elapsed Cycles [cycle]	4685154
L1/TEX Cache Throughput [%]	18.09	SM Active Cycles [cycle]	3708854.79
L2 Cache Throughput [%]	15.20	SM Frequency [Ghz]	1.39
DRAM Throughput [%]	0.71	DRAM Frequency [Ghz]	9.49

Small Grid This kernel grid is too small to fill the available resources on this device, resulting in only 0.78 full waves across all SMs. Look at Launch Statistics for more details.

Key Performance Indicators

Compute Throughput Breakdown

SM: Pipe Tensor Cycles Active [%]	34.95
SM: Issue Active [%]	11.80
SM: Inst Executed [%]	11.80
SM: Inst Executed Pipe Lsu [%]	9.86
SM: Mio2rf Writeback Active [%]	6.83
SM: Pipe Fma Cycles Active [%]	5.54
SM: Inst Executed Pipe Xu [%]	4.92
SM: Pipe Fmaheavy Cycles Active [%]	4.41
SM: Pipe Alu Cycles Active [%]	4.29
SM: Mio Inst Issued [%]	3.38
SM: Mio Pq Write Cycles Active [%]	0.56
SM: Mio Pq Read Cycles Active [%]	0.56
SM: Inst Executed Pipe Adu [%]	0.27
SM: Inst Executed Pipe Cbu Pred On Any [%]	0.00
SM: Inst Executed Pipe Uniform [%]	0.00
SM: Memory Throughput Internal Activity [%]	0
IDC: Request Cycles Active [%]	0
SM: Instruction Throughput Internal Activity [%]	0
SM: Inst Executed Pipe Tex [%]	0
SM: Pipe Fp64 Cycles Active [%]	0
SM: Inst Executed Pipe Ipa [%]	0

Memory Throughput Breakdown

L2: T Sectors [%]	15.20
L2: Lts2xbar Cycles Active [%]	15.14
L1: Data Pipe Lsu Wavefronts [%]	14.33
L1: Lsu Writeback Active [%]	13.66
L1: Lsuin Requests [%]	9.86
L1: M Xbar2l1tex Read Sectors [%]	8.77
L1: Data Bank Reads [%]	6.56
L2: D Sectors [%]	3.88
L2: Xbar2lts Cycles Active [%]	3.84
GPU: Compute Memory Access Throughput Internal Activity [%]	3.84
L2: T Tag Requests [%]	3.80
L1: M L1tex2xbar Req Cycles Active [%]	1.13
L1: Data Bank Writes [%]	1.11
DRAM: Cycles Active [%]	0.71
DRAM: Dram Sectors [%]	0.61
L2: D Sectors Fill Device [%]	0.25
L1: Texin Sm2tex Req Cycles Active [%]	0.03
L2: D Sectors Fill Sysmem [%]	0.00
L1: Tex Writeback Active [%]	0
L2: D Atomic Input Cycles Active [%]	0
L1: F Wavefronts [%]	0
L1: Data Pipe Tex Wavefronts [%]	0
GPU: Compute Memory Request Throughput Internal Activity [%]	0

Floating Point Operations Roofline
Performance [FLOP/s] (1 = 1e+12)
100 10 1 0.1
0.1 1 10 100 1,000 10,000
Arithmetic Intensity [FLOP/byte]
Floating Point Operations Roofline (Double Precision)
Performance [FLOP/s] (1 = 1e+12)
10 1 0.1
0.01 0.1 1 10 100
Arithmetic Intensity [FLOP/byte]
Floating Point Operations Roofline (Half Precision)
Performance [FLOP/s] (1 = 1e+13)
10 1
1 10 100 1,000 10,000
Arithmetic Intensity [FLOP/byte]
Floating Point Operations Roofline (Single Precision)
Performance [FLOP/s] (1 = 1e+12)
100 10 1 0.1
1 10 100 1,000 10,000 100,000 1e+06 1e+07
Arithmetic Intensity [FLOP/byte]
Tensor Core Operations Roofline
Performance [OP/s] (1 = 1e+14)
10 1 0.1
1 10 100 1,000 10,000 100,000 1e+06 1e+07 1e+08
Arithmetic Intensity [OP/byte]

	# Operations	# Operations / Cycle	# Operations / s	Peak %	eak Operations / Cycle	Peak Operations / s
Src:bf16 Dst:fp32 Sparsity:	137438953472	29349.13	40903671320546.27	69.91	41984	58512789318298.70
Src:bf16 Dst:fp32 Sparsity:	0	0	0	0	83968	117025578636597.41
Src:fp16 Dst:fp16 Sparsity:	0	0	0	0	83968	117025578636597.41
Src:fp16 Dst:fp16 Sparsity:	0	0	0	0	167936	234051157273194.81
Src:fp16 Dst:fp32 Sparsity:	0	0	0	0	41984	58512789318298.70
Src:fp16 Dst:fp32 Sparsity:	0	0	0	0	83968	117025578636597.41
Src:fp64	0	0	0	0	289.54	403536478057.23
Src:int1	0	0	0	0	1343488	1872409258185558...
Src:int4 Sparsity:off	0	0	0	0	335872	468102314546389.62
Src:int4 Sparsity:on	0	0	0	0	671744	936204629092779.25
Src:int8 Sparsity:off	0	0	0	0	167936	234051157273194.81
Src:int8 Sparsity:on	0	0	0	0	335872	468102314546389.62
Src:tf32 Dst:fp32 Sparsity:c	0	0	0	0	20992	29256394659149.35
Src:tf32 Dst:fp32 Sparsity:c	0	0	0	0	41984	58512789318298.70

PM Sampling

Timeline view of PM metrics sampled periodically over the workload duration. Data is collected across multiple passes. Use this section to understand how workload behavior changes over its runtime.

Maximum Sampling Interval [us]	2	# Pass Groups	2
Maximum Buffer Size [Mbyte]	21.95	-	-

0s
ms +0.5ms +1ms +1.5ms +2ms +2.5ms +3ms +3.5ms
Overview
Average Active Warps Per Cy 6.25 warp 0
Total Active Warps Per Cycle 512 warp 0
Blocks Launched 128 block 0
SM Active Cycles 2.81k cycle 0
Executed Ipc Active 0.8 inst/cycle 0
SM
SM ALU Pipe Throughput 100 % 0
SM FMA Light Pipe Throughp 100 % 0
SM FMA Heavy Pipe Through 100 % 0
SM Tensor Pipe Throughput 100 % 0
SM XU Pipe Throughput 100 % 0
DRAM
DRAM Throughput 100 % 0
DRAM Read Bandwidth 100 % 0
DRAM Write Bandwidth 100 % 0
L2 Cache
L2 Throughput 100 % 0
L2 Hit Rate 100 % 0
L1 Cache
L1 Throughput 100 % 0
L1 Writeback Throughput 545 cycle 0
L1 Hit Rate 100 % 0
L1 Wavefronts (Data) 571 0
Workload Execution
flash_fwd_kernel

Compute Workload Analysis

All

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	0.47	SM Busy [%]	34.95
Executed Ipc Active [inst/cycle]	0.60	Issue Slots Busy [%]	11.80
Issued Ipc Active [inst/cycle]	0.60		

Memory Workload Analysis

All

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with data for each memory unit.

Memory Throughput [Gbyte/s]	6.50	Mem Busy [%]	15.20
L1/TEX Hit Rate [%]	0.01	Max Bandwidth [%]	15.14
L2 Hit Rate [%]	98.37	Mem Pipes Busy [%]	9.86
L2 Compression Input Sectors [sector]	0	Local Memory Spilling Requests	1024
L2 Compression Ratio	0	Local Memory Spilling Request Overhead [%]	100
L2 Compression Success Rate [%]	0	L2 Persisting Size [Mbyte]	1.18

Local Memory Usage This workload accesses local memory. Local memory refers to a memory space that is private to each thread. Although it is logically distinct, local memory resides in global memory. This can lead to higher memory latency compared to the usage of registers or shared memory, a decrease in available bandwidth to global memory, and an increase in executed instructions.
10.16% of all local loads and 25.20% of all local stores spill into the L2 cache, accounting for 0.00% of all sectors requested in L2.
One common reason for the use of local memory is arrays being too large to fit into registers or not being indexed using compile time constants (see the "Local Memory" section of the CUDA Programming Guide. For more advice on how to reduce the usage of local memory refer to the documentation.

Key Performance Indicators

Shared Memory

	Instructions	Requests	Wavefronts	% Peak	Bank Conflicts
Shared Load	8192	8192	50366678	13.12	1543
Shared Load Matrix	12582912	12582912			
Shared Store	32768	32768	32768	0.00	0
Shared Store From Global Lo	2105344	2105344	8421376	0.55	0
Shared Atomic	0	0	0	0	0
Other	-	-	3805893	1.00	71
Total	14729216	14729216	62626715	14.66	1614

L1/TEX Cache

	Instructions	Requests	Wavefronts	% Peak	Sectors	Sectors/Req	Hit Rate	Bytes	Sector
Local Load	512	512	512	0.00	2048	4	89.84	65536	
Global Load	0	0	2105344	0.55	0	0	0	0	
Global Load To Shared Store (acce	0	0			0	0		0	
Global Load To Shared Store (bypa	2105344	2105344			33685504	16	-	1077936128	
Surface Load	0	0	0	0	0	0	0	0	
Texture Load	0	0	0	0	0	0	0	0	
Global Store	10240	10240	10858	0.00	133120	13	1.07	4259840	
Local Store	512	512	512	0.00	2048	4	74.80	65536	
Surface Store	0	0	0	0	0	0	0	0	
Global Reduction	0	0	0	0	0	0	0	0	
Surface Reduction	0	0	0	0	0	0	0	0	
Global Atomic ALU	0	0	0	0	0	0	0	0	
Global Atomic CAS	0								
Surface Atomic ALU	0	0	0	0	0	0	0	0	
Surface Atomic CAS	0								
Loads	2105856	2105856	2105856	0.55	33687552	16.00	0.01	1078001664	
Stores	10752	10752	11370	0.00	135168	12.57	2.19	4325376	
Atomics & Reductions	0	0	0	0	0	0	0	0	

L2 Cache

	Requests	Sectors	Sectors/Req	% Peak	Hit Rate	Bytes	Throughput	Sector Misses to Devic	ector Miss
L1/TEX Load	8421417	33685716	4.00	15.14	98.37	1077942912	320810232186.05	548400	
L1/TEX Store	33961	135168	3.98	0.06	100	4325376	1287289765.91	0	
L1/TEX Atomic ALU	0	0	0	0	0	0	0	0	
L1/TEX Atomic CAS	0	0	0	0	0	0	0	0	
L1/TEX Reduction	0	0	0	0	0	0	0	0	
L1/TEX Total	8455397	33820848	4.00	15.20	98.38	1082267136	322097179101.35	548896	
GCC Total	2016	8064	4	0.00	86.01	258048	76798537.17	1128	
ECC Total	-	0	-	0	-	0	0	0	

L2 Cache Eviction Policies

	First	Hit Rate	Last	Hit Rate	Normal	Hit Rate	Normal Demote	Hit Rate
L1/TEX Load	0	0	0	0	33685276	98.37	0	0
L1/TEX Store	0	0	0	0	135168	100	0	0
L1/TEX Atomic	0	0	0	0	0	0	-	-
L1/TEX Total	0	0	0	0	33818656	98.38	0	0
GPU Total	6075	95.44	0	0	33834938	98.37	0	0

Device Memory

	Sectors	% Peak	Bytes	Throughput
Load	550064	0.57	17602048	5238604978.95
Store	132788	0.14	4249216	1264623530.98
Total	682852	0.71	21851264	6503228509.93

Scheduler Statistics

Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions for. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler selects a single warp from which to issue one or more instructions (Issued Warp). On cycles with no eligible warps, the issue slot is skipped and no instruction is issued. Having many skipped issue slots indicates poor latency hiding.

Active Warps Per Scheduler [warp]	1.50	No Eligible [%]	85.10
Eligible Warps Per Scheduler [warp]	0.18	One or More Eligible [%]	14.90
Issued Warp Per Scheduler	0.15		

Warps Per Scheduler
GPU Maximum Warps Per Scheduler
Theoretical Warps Per Scheduler
Active Warps Per Scheduler
Eligible Warps Per Scheduler
Issued Warp Per Scheduler
0.0 4.0 8.0 12.0 16.0

Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.

Warp Cycles Per Issued Instruction [cycle]	10.09	Avg. Active Threads Per Warp	32.00
Warp Cycles Per Executed Instruction [cycle]	10.09	Avg. Not Predicated Off Threads Per Warp	28.38

Instruction Statistics

All

Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst]	181199360	Avg. Executed Instructions Per Scheduler [inst]	552437.07
Issued Instructions [inst]	181215839	Avg. Issued Instructions Per Scheduler [inst]	552487.31

NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Topology

The system does not have any NVLink connections.

NVLink Tables

Detailed tables with properties for each NVLink.

Logical NVLink Properties

The system does not have any NVLink connections.

NUMA Affinity

Non-uniform memory access (NUMA) affinities based on compute and memory distances for all GPUs.

NUMA ID Table

GPU ID	GPU Name	NUMA ID by CPU Affini	CPU Affinity	NUMA ID by Memory Affinity
0	NVIDIA GeForce RTX 3090	0	0-29	0

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	128	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	255	Static Shared Memory Per Block [byte/block]	0
Block Size	128	Dynamic Shared Memory Per Block [Kbyte/block]	49.15
Threads [thread]	16384	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	0.78	Shared Memory Configuration Size [Kbyte]	102.40
Uses Green Context	0	Stack Size	1024
# SMs [SM]	82	# TPCs	41
Enabled TPC IDs	all	-	-

Small Grid If you execute __syncthreads() to synchronize the threads of a block, it is recommended to have at least two blocks per multiprocessor (compared to the currently executed 1.6 blocks) This way, blocks that aren't waiting for __syncthreads() can keep the hardware busy.

Key Performance Indicators

Occupancy

All

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	16.67	Block Limit Registers [block]	2
Theoretical Active Warps per SM [warp]	8	Block Limit Shared Mem [block]	2
Achieved Occupancy [%]	12.53	Block Limit Warps [block]	12
Achieved Active Warps Per SM [warp]	6.01	Block Limit SM [block]	16

Theoretical Occupancy
Est. Local Speedup: 83.33%
The 2.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 12. This kernel's theoretical occupancy (16.7%) is limited by the number of required registers, and the required amount of shared memory.

Key Performance Indicators

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	3708854.79	Average L1 Active Cycles [cycle]	3708854.79
Average L2 Active Cycles [cycle]	1643759.98	Average SMSP Active Cycles [cycle]	3708139.52
Average DRAM Active Cycles [cycle]	227617.33	Total SM Elapsed Cycles [cycle]	383997494
Total L1 Elapsed Cycles [cycle]	383997494	Total L2 Elapsed Cycles [cycle]	222567504
Total SMSP Elapsed Cycles [cycle]	1535989976	Total DRAM Elapsed Cycles [cycle]	382740480

SMs Workload Imbalance
Est. Speedup: 16.41%
One or more SMs have a much lower number of active cycles than the average number of active cycles. Maximum instance value is 20.71% above the average, while the minimum instance value is 33.20% below the average.

Key Performance Indicators

SMSPs Workload Imbalance
Est. Speedup: 16.40%
One or more SMSPs have a much lower number of active cycles than the average number of active cycles. Maximum instance value is 20.71% above the average, while the minimum instance value is 33.18% below the average.

Key Performance Indicators

L1 Slices Workload Imbalance
Est. Speedup: 16.41%
One or more L1 Slices have a much lower number of active cycles than the average number of active cycles. Maximum instance value is 20.71% above the average, while the minimum instance value is 33.20% below the average.

Key Performance Indicators

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	3708854.79	2477538	4677806	304126093
SMSP Active Cycle	3708139.52	2477883	4676464	1216269761
L1 Active Cycles	3708854.79	2477538	4677806	304126093
L2 Active Cycles	1643759.98	1624811	1667280	78900479
DRAM Active Cycle	227617.33	226304	228672	2731408

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	796672	Branch Efficiency [%]	99.94
Branch Instructions Ratio [%]	0.00	Avg. Divergent Branches [branches]	1.56

Warp Stall Sampling (All Samples)

Location	Value	Value (%)
0x7fe0f5281870 in flash_fw...	8,343	6
0x7fe0f5281760 in flash_fw...	5,426	4
0x7fe0f5281e20 in flash_fw...	3,791	3
0x7fe0f52833c0 in flash_fwd...	1,706	1
0x7fe0f5281970 in flash_fw...	1,479	1

Most Instructions Executed

Location	Value	Value (%)
0x7fe0f5283fc0 in flash_fwd...	261,632	0
0x7fe0f5283fb0 in flash_fwd...	261,632	0
0x7fe0f5283fa0 in flash_fwd...	261,632	0
0x7fe0f5283f90 in flash_fwd...	261,632	0
0x7fe0f5283f80 in flash_fwd...	261,632	0

To customize your report even further, you might want to learn about custom sections and writing your own rules.
You might also want to consider adding individual metrics.